

```
$ python prompt_lab.py --show-comparison 3
```

```
=== BEFORE (Zero-Shot) ===
```

```
Prompt: "Explain this medical report to a patient in simple terms"
```

```
Output: "Hemoglobin low, WBC high, glucose high. Consult doctor."
```

```
→ clarity: 5/10 · too terse, no reassurance, no next step
```

```
=== AFTER (Chain-of-Thought + Role Prompting) ===
```

```
Prompt: "You are a caring family doctor... think step-by-step..."
```

```
Output:
```

```
"Your hemoglobin is a little low, which can sometimes cause  
tiredness – this is often related to iron levels. Your white  
blood cell count is slightly elevated, which can simply mean  
your body is fighting off a minor infection. Your fasting  
glucose is a bit high, so it may help to mention this to your  
doctor at your next visit. Nothing here looks alarming, but  
it's worth a short follow-up."
```

```
→ clarity: 9/10 · warm tone, explains cause, gives next step
```

```
$ |
```

somit@runqta-lab — python — 92x30

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```
$ |
```



Medical Report Explainer

localhost:8501

Medical Report Explainer

MODEL

Groq · llama-3.3-70b-versatile ▼

PROMPT TEMPLATE

Role + Chain-of-Thought ▼

READING LEVEL

Grade 6 (Simple) ▼

Explain Report

Clear

Medical Report Explainer

Paste your lab report below and get a plain-language explanation.

YOUR REPORT TEXT

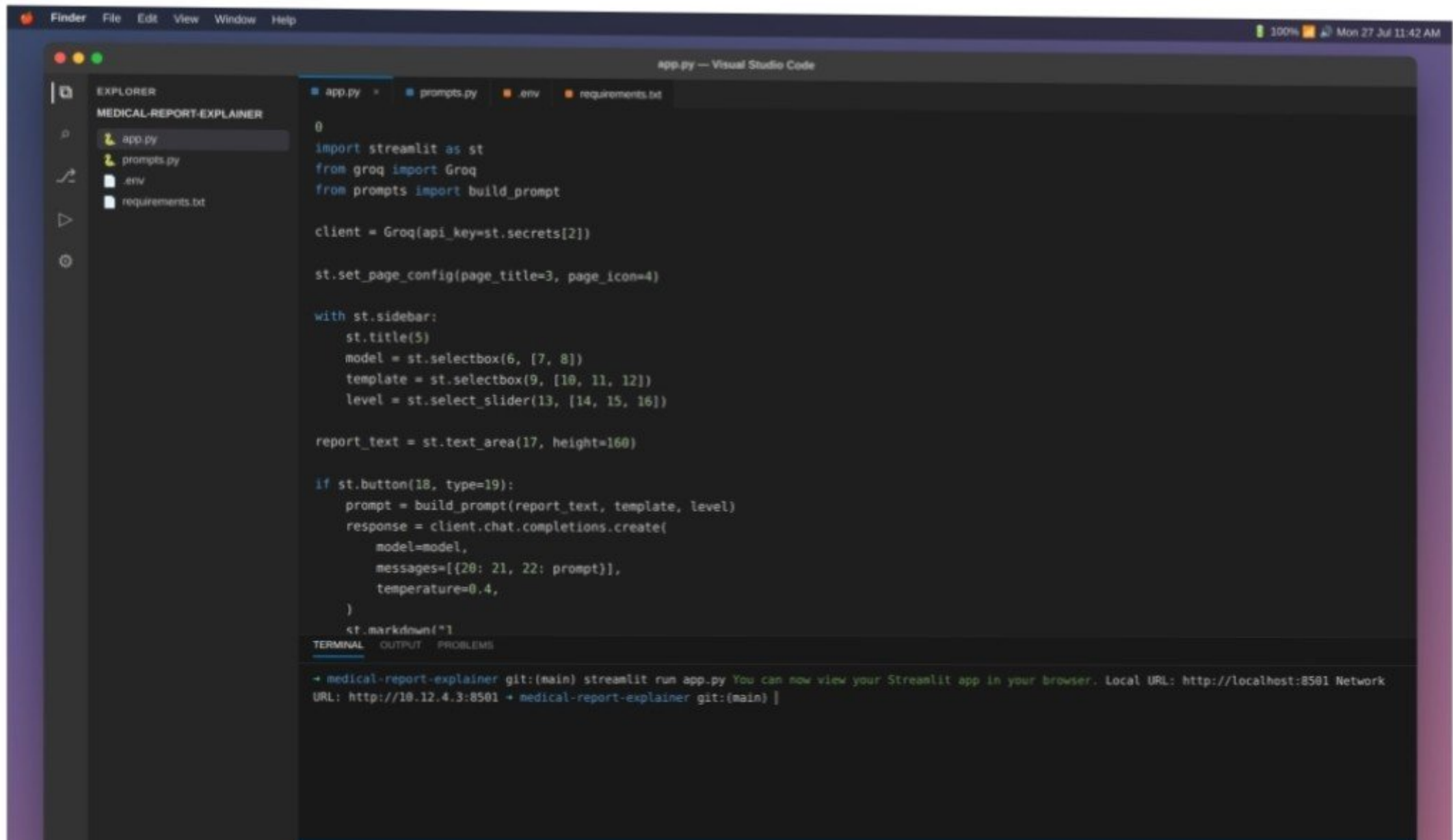
Hemoglobin: 10.2 g/dL (Low)
WBC Count: 11,800 /uL (High)
Glucose (Fasting): 118 mg/dL (High)

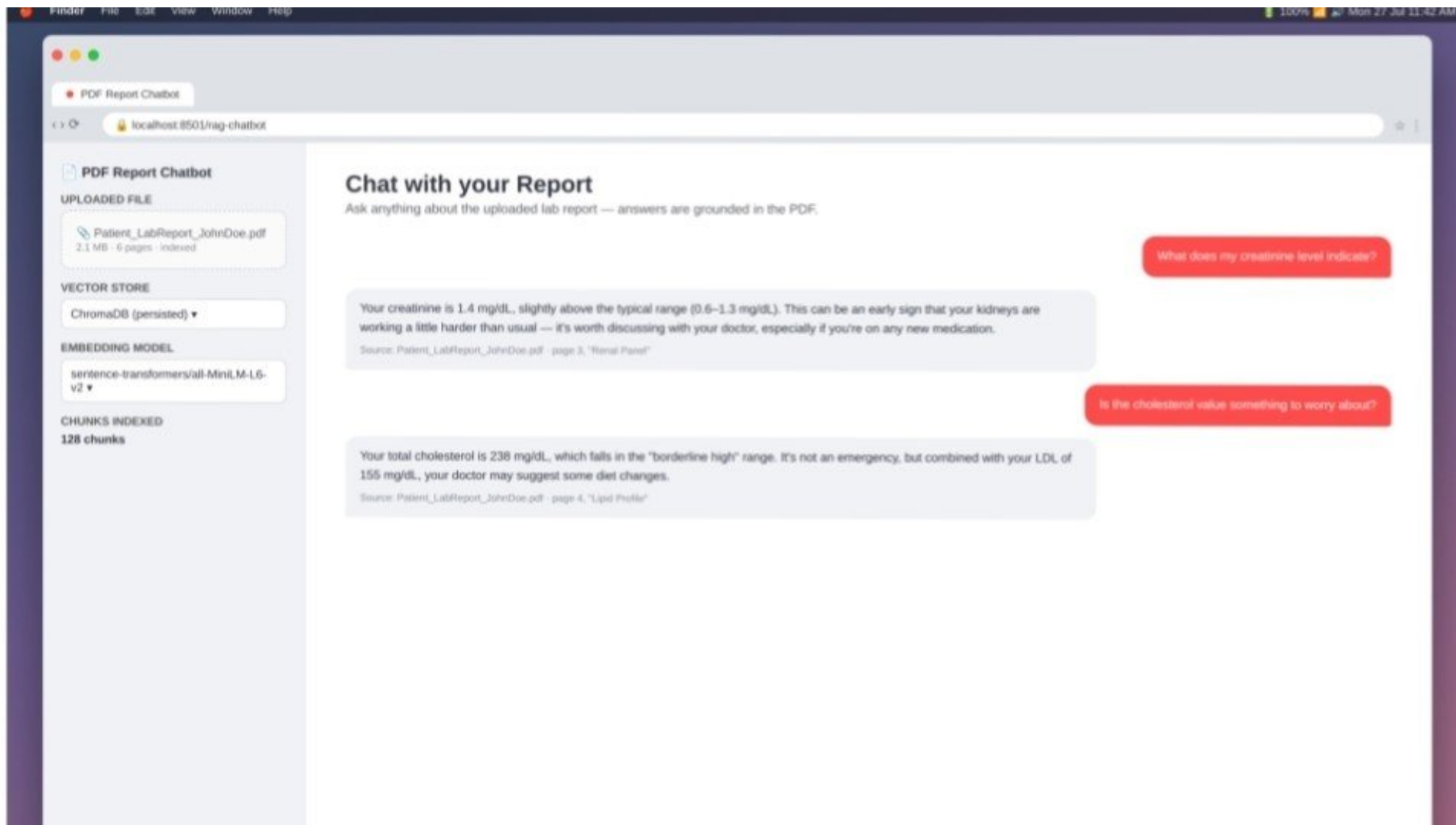
EXPLANATION

Your hemoglobin is a touch low, which can sometimes make you feel tired — it's often related to iron. Your white blood cell count is a little high, which us...

body is fighting something minor. Your fasting glucose is slightly above normal, so it's worth mentioning at your next check-up. Nothing here looks urgent.

4/15





```
$ python ingest.py --file Patient_LabReport_JohnDoe.pdf
```

[OK] Extracted 9.842 KB

[OK] Extracted 9,842 characters

```
Splitting text (chunk_size=500, overlap=50)...
```

```
[OK] Created 128 chunks
```

```

Loading embedding model: all-MiniLM-L6-v2 ...
[OK] Model loaded.

```

```
[OK] Model loaded on device: cpu
```

Generating embeddings...

```
100% ██████████ 128/128 [00:07<00:00, 17.4it/s]
[OK] Persisted 128 vectors to /home/.../vectors.npy
```

```
[OK] Persisted 128 vectors to ./chroma_db (collection: lab_reports)
Running sanity query: "createIndex lab_reports"
```

```
Running sanity query: "creatinine level"
```

```
[OK] top-3 matches retrieved - avg. similarity 0.87
```



```
Finder  File  Edit  View  Window  Help

somil@runqta-lab — python agent.py — 100x34

$ python agent.py --report sample_report.txt

> Entering new AgentExecutor chain...

Thought: I should first extract each lab value and flag which are
abnormal before explaining anything to the patient.
Action: ExtractLabValues
Action Input: "sample_report.txt"
Observation: {'Hemoglobin': 10.2, 'WBC': 11800, 'Glucose': 118}

Thought: Now I should check which values are outside normal range.
Action: RiskFlagger
Action Input: {'Hemoglobin': 10.2, 'WBC': 11800, 'Glucose': 118}
Observation: flags=['Hemoglobin: Low', 'WBC: High', 'Glucose: High'], risk='moderate'

Thought: I should look up simple definitions for these terms for
the patient before writing the final explanation.
Action: MedicalTerminologyLookup
Action Input: "creatinine, hemoglobin, WBC"
Observation: 3 definitions retrieved from glossary.json

Thought: I now have enough information to answer.
Final Answer: Your hemoglobin is a little low, your white blood
cell count is a bit high, and your fasting glucose is slightly elevated.
None of this looks urgent, but it's worth a follow-up.

> Finished chain.
[memory] saved turn to ConversationBufferMemory (session: 8f21c)

❗
```

EXPLORER

MEDICAL-REPORT-EXPLAINER

agent.py
tools.py
memory.py
glossary.json

```
0  
from langchain.agents import initialize_agent, AgentType  
from langchain.memory import ConversationBufferMemory  
from langchain_groq import ChatGroq  
from tools import ExtractLabValues, RiskFlagger, MedicalTerminologyLookup  
  
llm = ChatGroq(model='llm', temperature=0.2)  
  
memory = ConversationBufferMemory(memory_key='2', return_messages=True)  
  
tools = [ExtractLabValues(), RiskFlagger(), MedicalTerminologyLookup()]  
  
agent = initialize_agent(  
    tools=tools,  
    llm=llm,  
    agent=AgentType.CONVERSATIONAL_REACT_DESCRIPTION,  
    memory=memory,  
    verbose=True,  
    max_iterations=6,  
)  
  
if __name__ == '__main__':  
    result = agent.run('4')  
    print(result)
```

TERMINAL

```
+ medical-report-explainer git:(main) python agent.py --report sample_report.txt [OK] agent finished in 4 steps · 3 tool calls · memory updated + medical-  
report-explainer git:(main) |
```


Medical Report Explorer

localhost:8503/tools

Medical Report Explorer

CONNECTED TOOLS

☒ Gmail API☒ Google Calendar API

DOCTOR'S EMAIL

dr.mehta@carehospital.in

Share & Follow Up

Send this explanation to your doctor or schedule a follow-up — no copy-paste needed.

✉ EMAIL SUMMARY TO DOCTOR

Sends the plain-language explanation + flagged values as a PDF attachment.

Send to Dr. Mehta

📅 SCHEDULE FOLLOW-UP

Books a 20-minute slot on your Google Calendar based on your doctor's availability.

Schedule Follow-up

```
$ python tools.py --action send-summary --to dr.mehta@carehospital.in
```

```
[gmail_tool] building MIME message with PDF attachment...
```

```
[gmail_tool] POST https://gmail.googleapis.com/gmail/v1/users/me/messages/send
```

```
[OK] 200 <Response> message_id="18f2a9c6e2d4"
```

```
[OK] Email sent to dr.mehta@carehospital.in
```

```
$ python tools.py --action schedule-followup --slot "2026-08-05T10:30:00+05:30"
```

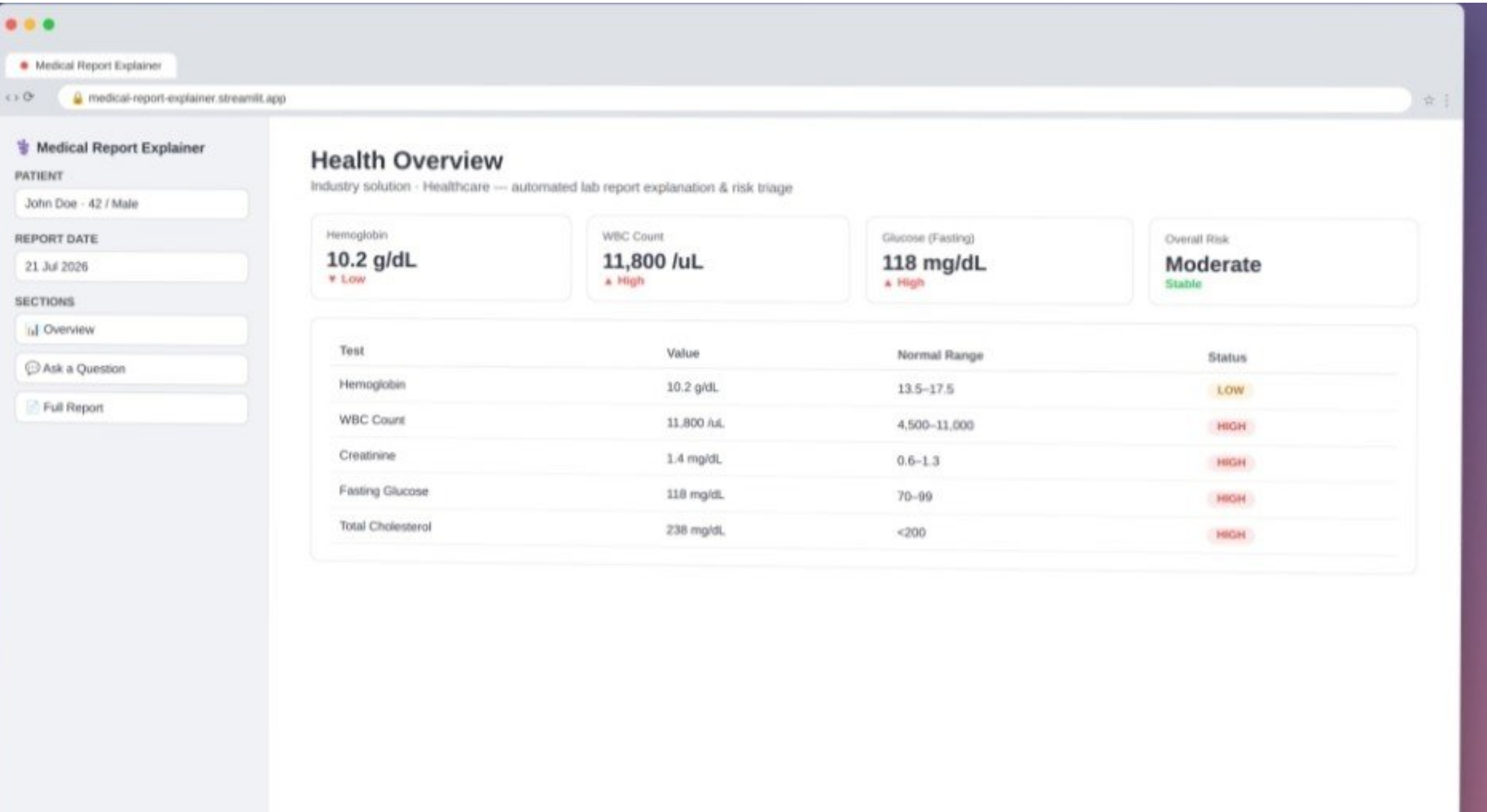
```
[calendar_tool] POST https://www.googleapis.com/calendar/v3/calendars/primary/events
```

```
[OK] 200 <Response>
```

```
{
  "status": "confirmed",
  "summary": "Follow-up: Lab Report Review",
  "start": "2026-08-05T10:30:00+05:30",
  "end": "2026-08-05T10:50:00+05:30",
  "attendees": ["dr.mehta@carehospital.in"]
}
```

```
[OK] Event created and added to Google Calendar
```

```
$ |
```





Ask a Question

medical-report-explainer.streamlit.app/ask

Medical Report Explainer

SECTIONS

Overview

Ask a Question

Full Report

Download Explanation (PDF)

Ask a Question

Grounded in your uploaded report · powered by RAG + Groq

Should I be worried about the cholesterol and glucose together?

Not urgently — but together, slightly high cholesterol and glucose can be an early signal worth watching. Many people improve both with small changes to diet and activity. Your doctor may suggest a repeat test in 3 months.

Sources: page 2 "Lipid Profile", page 3 "Metabolic Panel"

Type your question about this report...

Ask

medical-report-explainer

Deployed from GitHub · somiljain/medical-report-explainer · branch: main

Status

● **Running**

Last Deploy

2 min ago

Python

3.11

Resource Usage

312 MB / 1 GB

APP URL

<https://medical-report-explainer.streamlit.app>

DEPLOY LOG

```
[11:40:02] Cloning repository...  
[11:40:05] Installing requirements.txt (14 packages)...  
[11:40:38] Starting app server...  
[11:40:41] ✓ App is live
```

EXPLORER

MEDICAL-REPORT-EXPLAINER

prompt_lab.py
sample_report.txt
requirements.txt

```
0  
1  
  
REPORT_SNIPPET = 2  
  
def class="fn">zero_shot_prompt(text):  
    return f3  
  
def class="fn">few_shot_prompt(text):  
    examples = 4  
    return f5  
  
def class="fn">cot_prompt(text):  
    return f6  
  
def class="fn">role_prompt(text):  
    return f7  
  
TECHNIQUES = {  
    8: zero_shot_prompt,  
    9: few_shot_prompt,  
    10: cot_prompt,  
    11: role_prompt,  
}
```

TERMINAL

```
+ medical-report-explainer git:(main) python prompt_lab.py --technique cot --run all [OK] Loaded 20 prompts across 4 techniques [OK] zero_shot → avg. clarity score: 6.1/10 [OK] few_shot → avg. clarity score: 7.4/10 [OK] chain_of_thought → avg. clarity score: 8.8/10 [OK] role_prompt → avg. clarity score: 8.3/10 [NOTE] Saving before/after pairs → ./outputs/prompt_comparisons.json + medical-report-explainer git:(main) |
```